Alex Arkilanian

Education

PhD in Biology (2022-2026)
Memorial University of Newfoundland, St. John's, Newfoundland and Labrador, Canada

Msc in Biology (2018-2021)
McGill University, Montréal, Québec, Canada

BSc with Honours (2015-2018)
McGill University, Montréal, Québec, Canada

Professional experience

Founder and chief analyst (2022 - Present)
Freeflow Nature Consulting Group (freeflownature.ca)

- Founded an ecological consulting company focused on watershed conservation concerning fish migration and the remediation of aquatic barriers such as dams and culverts.
- Designed an open-source software solution to measure the ecological connectivity of river networks with the aim of prioritizing aquatic barriers (dams, culverts) for removal or remediation.
- Secured contracts with Environment and Climate Change Canada (ECCC) and the Department of Fisheries and Oceans (DFO) to incorporate invasive species risk into the measurement of river connectivity.

Software Consultant (2022 - 2023)
DFO (Department of Fisheries and Oceans) and MCK (Mohawk Council of Kahnawá:ke)

- Developed a multi-risk ranking system for aquatic barriers to prioritize the most significant barriers given a range of objectives.
- Collaborated with the local indigenous council (MCK) to incorporate cultural value of streams into an assessment of priority for removal of aquatic barriers.
- Led trainings with DFO staff in Quebec on the use of these new ranking workflow and analyses.

Conservation Consultant (2020 - 2022)
MELCC (Quebec environment ministry), MFFP (Quebec fauna and flora ministry), and MTQ (Quebec transportation ministry)

- Designed a workflow to assess meaningful barriers to individual white sucker (*Catastomus comersonii)* populations given spawning habitat and migration needs.
- Designed a community-level fish connectivity analysis for Quebec watersheds with recommendations on conservation priorities to conserve fish biodiversity (~100 species).

- Consulted with Quebec's top interdisciplinary scientists across 3 ministries to align project with government priorities and the larger St. Lawrence Action Plan (2011-2016).

Scientific Consultant (2019 - 2020)
MFFP (Quebec fauna and flora ministry) and Nature United

- Led a report describing and guiding future connectivity conservation in New England and the Eastern Canadian provinces
- Managed a team of 5 students in literature review, statistical meta-analyses, and the production of a final report
- Presented final report to a team of New England's state conservation authorities and Eastern Canadian provinces' conservation ministries.

Teaching experience

Teaching assistant (2022 - Present)
Memorial University of Newfoundland

- Designed analysis workflows and R resources for student labs in Conservation Biology I (BIOL 3650) and Community and Ecosystem Ecology (BIOL 4360).
- Led analysis labs in Community and Ecosystem Ecology (BIOL 4360)

Instructor (2019 - 2021)
Computational Biodiversity Science and Services (BIOS^2)

- Led training of over 40 students on the use of git version control, bash programming, R programming, and data visualization.

Teaching assistant (2018 - 2021)
McGill University

- Helped transition Biology of Organisms (BIOL 101) lectures to an engaging fully online class during the COVID-19 pandemic.
- Designed and delivered lecture content on ecological network for the Biodiversity and Ecosystems course (BIOL 310).
- Led pre-exam seminars in Introduction to Ecology and Evolution (BIOL 215) summarizing lecture content and responding to student questions.

Volunteer experience

Chair (2024 - Present)
Aquatic Conservation Initiative

- Leading Board of Director meetings
- Liaising with field technicians and staff to ensure field work proceeds safely and successfully.
- Developing the short and long term visions of the organization in line with its mission and objectives

Graduate student representative (2024 - Present)
MUN Board of Regents

- Representing graduate student interests on the Board of Regents

Seminar & professional development coordinator (2023 - 2024)
MUN Biology Graduate Student Association

- Identified speakers for the weekly seminar series and co-organized seminars. Contributed to the increasing success of the seminar series following low attendance immediately following the university closure due to the COVID-19 pandemic.
- Organized and helped develop workshops to train graduate students in important professional skills: git training and time management.
- Organizing the annual Biology Graduate Student Symposium (2024) which saw over 100 attendants and over 30 presenters.

Director of Create platform (2018 - 2020)
SciComm Collective

- Identified authors to generate articles and blog posts for SciComm Collective (over 10 articles published)
- Developed a network of scientific illustrators to partner with scientists looking to create complex visualization for articles or presentation.

Head cook (2018 - 2019)
Mile End Mission

- Organized and prepared Thursday lunches for the community

Volunteer (2018 - 2020)
Consent McGill

- Provided education and outreach within McGill, organized and ran events, and produced media for distribution (podcasts, flyers, infographics)

Awards and prizes

Canada Graduate Scholarship (nominated to final round; $150,000)
(2024) National Sciences and Engineering Research Council of Canada

Vanier Canada Graduate Scholarship (nominated to final round; $105,000)
(2024) National Sciences and Engineering Research Council of Canada

Best PhD Presentation (runner-up)
(2023) MUN Biology Graduate Student Symposium, St. John's, NL, Canada

Gordon Edwards Graduate Award ($1,000)
(2021) McGill University

Graduate Excellence Award ($2,500)
(2019) McGill University

Best poster (runner-up)
(2019) Quebec Center for Biodiversity Science conference, Montreal, QC, Canada

Computational Biodiversity Science and Services (BIOS^2) Fellowship ($43,000)
(2019) National Sciences and Engineering Research Council of Canada

Alexandra Irwin Cowie Fellowship ($2,500)
(2018) McGill University

UZH Biology Undergraduate Summer School ($3,000)
(2017) University of Zurich

EcoLac Fellowship ($650)
(2017) National Sciences and Engineering Research Council of Canada

Publications

Journal articles

1. Arkilanian, A., & Leroux, S., 2024. *Impacts of altered salmonid spawning timing on headwater ecosystems*. (in prep)
2. Paquette, C., Gagné, S., Gaudet-Boulay, M., Ouellet-Cauchon, G., Leclerc, V., **Arkilanian, A.**, Taranu, E. Z., Beisner, E. B. & Fugère, V. 2024. *Drivers of walleye, brook trout and lake trout abundance in Quebec lakes.* Canadian Journal of Fisheries and Aquatic Sciences (planned submission).
3. Wiersma, F. Y., Mahlum, S., Oldford, G., **Arkilanian, A.**, Kehler, D. & Cote D. 2024. *Does including a fish's perspective improve river connectivity metrics?* River Research and Applications (planned submission).
4. Arkilanian, A. A., Clements, C. F., Ozgul, A., Baruah, G., 2020. *Effect of time-series length and resolution on abundance- and trait-based early warning signals of population declines*. Ecology, 101(7), e03040.

Reports

1. Arkilanian, A. 2023. *River Connectivity Analysis of the Chateauguay watershed.* (Report delivered to the Department of Fisheries and Oceans).
2. Arkilanian, A. 2022. *dci: an R package to calculate the dendritic connectivity index in river networks.*
3. Arkilanian, A. & Gonzalez A., 2022. *Multi-species aquatic connectivity assessment for the Yamaska and Saint-Charles watersheds*. (Report presented to the Ministère de l'Environnement et la Lutte contre les Changements Climatiques).
4. Arkilanian, A. & Gonzalez A., 2021. *Aquatic connectivity assessment for the Yamaska watershed: the case of the white sucker*. (Report presented to the Ministère de l'Environnement et la Lutte contre les Changements Climatiques).

5. Arkilanian, A., Larocque, G. Lucet, V, Schrock, D. Denépoux, C. and Gonzalez, A., 2020. *A review of ecological connectivity science in the Region of Resolution 40-3*. (Report presented to the Ministère de la faune, de la forêt et des parcs du Québec for the New England Governors and Eastern Canadian Premiers working group on ecological connectivity). 79 pages.

Theses

1. Arkilanian, A. & Gonzalez A., 2021. *The impact of habitat quality and anthropogenic fragmentation on the functional connectivity of river networks for fish*. McGill University, Master's Thesis
2. Arkilanian, A., Stuart-Fox, D. & Hendry A., 2018. *An Evaluation of Colour Patterns with Patch Adjacency in the Trinidadian Guppy (Poecilia reticulata).* McGill University, Bachelor's Honours Thesis

Newspaper articles

1. Arkilanian, A. 2019. *Mapping nature's contributions to people*. McGill Tribune.
2. Arkilanian, A. 2019. *Leaping into the sixth mass extinction*. McGill Tribune.
3. Arkilanian, A. 2019. *The devastating impacts of human conflict on the natural world*. SciComm Collective.

Presentations

1. Arkilanian, A. 2023. *Assessing fragmentation in the rivers of the St. Lawrence Lowlands: a partnership with Quebec's environment and transport ministries.* Biology Graduate Student Symposium, St. John's, NL, Canada
2. Arkilanian, A. 2023. *A closer look into our rivers.* Kidsfest, St. John's, NL, Canada
3. Arkilanian, A. 2019. *Ecological networks.* Invited lecture in Biodiversity and Ecosystems undergraduate class at McGill University, Montreal, Quebec, Canada
4. Arkilanian, A. 2019. *Towards finding Earth's heartbeat and keeping our finger on the pulse*. Pint of Science, Montreal, Quebec, Canada
5. Arkilanian, A. 2019. Poster Presentation: *A connectivity analysis of the aquatic ecosystems of the Saint-Lawrence Lowlands*. Quebec Centre for Biodiversity Science conference, Montreal, Quebec, Canada
6. Arkilanian, A. 2018. *The effect of relative sampling time series length and resolution on early warning signals of population collapse*. Quebec Centre for Biodiversity Science conference, Montreal, Quebec, Canada
7. Arkilanian, A. 2017. *Context-dependent temporal resolution in detecting early warning signals of population extinction.* Biology Undergraduate Summer School conference, Switzerland